Class 12 Mathematics

Chapter 11: Three Dimensional Geometry

Practice Set B: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ Board-pattern — 1 mark

The lines $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z}{3}$ and $\frac{x-3}{-4}=\frac{y+2}{2}=\frac{z-3}{-6}$ are:

(A) parallel but not coincident

(B) **coincident**

(C) intersecting at exactly one point

(D) skew

Formula / idea: Lines with proportional DRs are parallel; if they also share a point, they coincide.

1 Compare the DRs $2,-1,3$ and $-4,2,-6$.
$\frac{-4}{2}=\frac{2}{-1}=\frac{-6}{3}=-2 \Rightarrow$ the lines are parallel

2 Check whether the point $(3,-2,3)$ of the second line lies on the first. ✓ 1
$\frac{3-1}{2}=\frac{-2+1}{-1}=\frac{3}{3}=1$ ✓

3 Parallel lines with a common point coincide. Hence the answer is option (B).

Answer **(B) coincident**

Watch out: Choosing "parallel but not coincident" (option A) without checking for a common point.

2 MCQ NCERT Exemplar 1 mark

A point A in space is such that $\overrightarrow{OA}$ is inclined at 60° to OX and at 45° to OY, and $|\overrightarrow{OA}| = 10$ units. The position vector of A is:

(A) $5\hat{i} + 5\sqrt{2}\hat{j} \pm 5\hat{k}$ **(B)** $5\sqrt{3}\hat{i} + 5\sqrt{2}\hat{j} \pm 5\hat{k}$

(C) $5\hat{i} + 5\sqrt{2}\hat{j} \pm 5\sqrt{2}\hat{k}$ **(D)** $\frac{1}{2}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} \pm \frac{1}{2}\hat{k}$

Formula / idea: A vector of length r with direction cosines l, m, n is $r(l\hat{i} + m\hat{j} + n\hat{k})$.

1 Write the two known direction cosines.
$l = \cos 60^\circ = \frac{1}{2}, \quad m = \cos 45^\circ = \frac{1}{\sqrt{2}}$

2 Use $l^2 + m^2 + n^2 = 1$ to find n.
$n^2 = 1 - \frac{1}{4} - \frac{1}{2} = \frac{1}{4} \Rightarrow n = \pm\frac{1}{2}$

3 Multiply the direction cosines by $|\overrightarrow{OA}| = 10$. ✓ 1
$\overrightarrow{OA} = 10\left(\frac{1}{2}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} \pm \frac{1}{2}\hat{k}\right) = 5\hat{i} + 5\sqrt{2}\hat{j} \pm 5\hat{k}$

4 Hence the answer is option (A).

Answer **(A)** $5\hat{i} + 5\sqrt{2}\hat{j} \pm 5\hat{k}$

Watch out: Stopping at the direction cosines (option D) and forgetting to multiply by the length 10.

3 MCQ CBSE 2023 1 mark

Equation of a line passing through the point $(1, 2, 3)$ and equally inclined to the coordinate axes is:

(A) $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ **(B)** $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$

(C) $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-1}{3}$ **(D)** $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1}$

Formula / idea: A line equally inclined to the axes has direction cosines $l = m = n$, so DRs $1, 1, 1$.

1 Equal angles with the axes give equal direction cosines.
$l = m = n \Rightarrow$ DRs $1, 1, 1$

2 Write the line through $(1, 2, 3)$ with DRs $1, 1, 1$. ✓ 1
$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1}$

3 Hence the answer is option (D).

Answer **(D)** $\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-3}{1}$

Watch out: Taking the point's coordinates $1, 2, 3$ as DRs (options A and B); the DRs must be equal.

4 MCQ NCERT Misc. Ex. Q1 1 mark

The angle between the lines whose direction ratios are a, b, c and $b-c,\ c-a,\ a-b$ is:

(A) 0° **(B)** 30° **(C)** 60° **(D)** 90°

Formula / idea: Lines with DRs a_1, b_1, c_1 and a_2, b_2, c_2 are perpendicular if $a_1a_2+b_1b_2+c_1c_2=0$.

1 Find $a_1a_2+b_1b_2+c_1c_2$.
$a(b-c)+b(c-a)+c(a-b)$

2 Expand; every term cancels. ✓ 1
$=ab-ac+bc-ab+ca-bc=0$

3 So $\cos\theta=0$ and $\theta=90^\circ$. Answer: option (D).

Answer **(D)** 90°

Watch out: Guessing 0° (option A) because both sets of DRs use the same letters.

5 MCQ Board-pattern 1 mark

The lines $\frac{2x-1}{4}=\frac{1-y}{k}=\frac{z+2}{3}$ and $\frac{x-3}{k}=y+1=\frac{2-z}{2}$ are perpendicular to each other. The value of k is:

(A) 2 **(B)** -6 **(C)** 6 **(D)** -2

Formula / idea: Put both lines in the form $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$ first; then $a_1a_2+b_1b_2+c_1c_2=0$.

1 Make the coefficients of x, y, z equal to $+1$ in the first line.
$\frac{x-\frac{1}{2}}{2}=\frac{y-1}{-k}=\frac{z+2}{3}\ \Rightarrow\ \text{DRs } 2, -k, 3$

2 Do the same for the second line.
$\frac{x-3}{k}=\frac{y+1}{1}=\frac{z-2}{-2}\ \Rightarrow\ \text{DRs } k, 1, -2$

3 Apply the perpendicular condition. ✓ 1
$2k-k-6=0\ \Rightarrow\ k=6$

4	Hence the answer is option (C).	

Answer **(C)** 6

Watch out: Reading DRs straight off the given form: keeping 4 for $2x$ gives $k = 2$; missing the sign of $2 - z$ gives $k = -6$.

6 **Assertion-Reason** CBSE SQP 2022-23 1 mark

Assertion (A): The acute angle between the line $\vec{r} = \hat{i} + \hat{j} + 2\hat{k} + \lambda(\hat{i} - \hat{j})$ and the x-axis is $\frac{\pi}{4}$. **Reason (R):** The acute angle θ between the lines $\vec{r} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k} + \lambda(a_1\hat{i} + b_1\hat{j} + c_1\hat{k})$ and $\vec{r} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k} + \mu(a_2\hat{i} + b_2\hat{j} + c_2\hat{k})$ is given by $\cos\theta = \dfrac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: The x-axis is the line $\vec{r} = t\hat{i}$, with direction $\hat{i}$.

1	Check Assertion: take $\vec{b}_1 = \hat{i} - \hat{j}$ and $\vec{b}_2 = \hat{i}$ for the x-axis. $\cos\theta = \dfrac{|1(1) + (-1)(0) + 0(0)|}{\sqrt{2}\cdot 1} = \dfrac{1}{\sqrt{2}} \Rightarrow \theta = \dfrac{\pi}{4}$	
2	So A is true.	
3	Check Reason: it is the standard formula for the acute angle. So R is true.	
4	A is obtained by applying exactly this formula, so R explains A.	✓ 1
5	Hence the answer is option (A).	

Answer **(A) Both true; R is the correct explanation of A.**

Watch out: Using the point $\hat{i} + \hat{j} + 2\hat{k}$ instead of the direction $\hat{i} - \hat{j}$ to find the angle.

7 **Short answer** PYQ-style 2 marks

Find the vector and the Cartesian equations of the line passing through the points $A(0, 5, -2)$ and $B(3, -1, 2)$. Hence find the coordinates of the point on this line whose x-coordinate is 6.

Formula / idea: Line through $\vec{a}$ and $\vec{b}$: $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

1	Find the direction $\vec{b} - \vec{a}$. $(3\hat{i} - \hat{j} + 2\hat{k}) - (5\hat{j} - 2\hat{k}) = 3\hat{i} - 6\hat{j} + 4\hat{k}$	✓ ½

2	Write the vector equation. $\vec{r} = (5\hat{j} - 2\hat{k}) + \lambda(3\hat{i} - 6\hat{j} + 4\hat{k})$	✓ ½
3	Write the Cartesian equation through $A(0, 5, -2)$. $\frac{x-0}{3} = \frac{y-5}{-6} = \frac{z+2}{4}$	✓ ½
4	A general point is $(3\lambda, 5 - 6\lambda, 4\lambda - 2)$; put $3\lambda = 6$. $\lambda = 2 \Rightarrow (6, -7, 6)$	✓ ½

Answer $\vec{r} = (5\hat{j} - 2\hat{k}) + \lambda(3\hat{i} - 6\hat{j} + 4\hat{k})$; $\frac{x}{3} = \frac{y-5}{-6} = \frac{z+2}{4}$; **point** $(6, -7, 6)$

Watch out: Taking the direction as $3\hat{i} - \hat{j} + 2\hat{k}$, the position vector of B, instead of $\vec{b} - \vec{a}$.

Level 2

16 marks

8 **Short answer** NCERT Exemplar 2 marks

Prove that the lines $x = py + q,\ z = ry + s$ and $x = p'y + q',\ z = r'y + s'$ are perpendicular if $pp' + rr' + 1 = 0$.

Formula / idea: Perpendicular lines: $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

1	Make y the subject in the first line and equate. $\frac{x-q}{p} = \frac{y-0}{1} = \frac{z-s}{r} \Rightarrow \text{DRs } p, 1, r$	✓ ½
2	Do the same for the second line. $\frac{x-q'}{p'} = \frac{y-0}{1} = \frac{z-s'}{r'} \Rightarrow \text{DRs } p', 1, r'$	✓ ½
3	The lines are perpendicular if the sum of products of DRs is zero. $pp' + 1 \cdot 1 + rr' = 0$	✓ ½
4	Hence the lines are perpendicular if $pp' + rr' + 1 = 0$.	✓ ½

Answer **DRs are** $p, 1, r$ **and** $p', 1, r'$**; perpendicular when** $pp' + 1 + rr' = 0$**.**

Watch out: Taking DRs p, q, r; here q and s fix a point, and the DR of y is 1.

9 **Short answer** CBSE 2025 3 marks

Verify that the lines given by $\vec{r} = (1-\lambda)\hat{i} + (\lambda-2)\hat{j} + (3-2\lambda)\hat{k}$ and $\vec{r} = (\mu+1)\hat{i} + (2\mu-1)\hat{j} - (2\mu+1)\hat{k}$ are skew lines. Hence, find the shortest distance between the lines.

Formula / idea: $d = \left|\dfrac{(\vec{b}_1 \times \vec{b}_2)\cdot(\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$; lines are skew if not parallel and $d \neq 0$.

1 Separate the constant and parameter parts of each line. ✓ ½

$\vec{a}_1 = \hat{i} - 2\hat{j} + 3\hat{k}, \quad \vec{b}_1 = -\hat{i} + \hat{j} - 2\hat{k}$

$\vec{a}_2 = \hat{i} - \hat{j} - \hat{k}, \quad \vec{b}_2 = \hat{i} + 2\hat{j} - 2\hat{k}$

2 Find $\vec{b}_1 \times \vec{b}_2$. ✓ 1

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 1 & 2 & -2 \end{vmatrix} = 2\hat{i} - 4\hat{j} - 3\hat{k}$$

Non-zero, so the lines are not parallel.

3 Find $\vec{a}_2 - \vec{a}_1$ and the dot product. ✓ ½

$\vec{a}_2 - \vec{a}_1 = \hat{j} - 4\hat{k}, \quad (2)(0) + (-4)(1) + (-3)(-4) = 8$

4 Put into the formula. ✓ ½

$$d = \frac{|8|}{\sqrt{4+16+9}} = \frac{8}{\sqrt{29}} \text{ units}$$

5 Since $d \neq 0$, the lines do not meet; they are not parallel either. Hence they are skew. ✓ ½

Answer **Skew lines; shortest distance $= \dfrac{8}{\sqrt{29}}$ units**

Watch out: Reading $\vec{b}_1$ as $\hat{i} + \hat{j} + 2\hat{k}$: the coefficients of λ in $1-\lambda$ and $3-2\lambda$ are negative.

10 **Short answer** Board-pattern 3 marks

Show that the lines $\dfrac{x-1}{2} = \dfrac{y+1}{-3} = \dfrac{z}{6}$ and $\dfrac{2x-6}{4} = -\dfrac{y}{3} = \dfrac{2z-2}{12}$ are parallel, and find the distance between them.

Formula / idea: Parallel lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$, $\vec{r} = \vec{a}_2 + \mu\vec{b}$: $d = \dfrac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$.

1 Write the second line in standard form. ✓ ½

$$\frac{x-3}{2} = \frac{y-0}{-3} = \frac{z-1}{6}$$

2 Both lines have DRs $2, -3, 6$, so they are parallel. Read the points. ✓ ½

$\vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}, \quad \vec{a}_1 = \hat{i} - \hat{j}, \quad \vec{a}_2 = 3\hat{i} + \hat{k}$

3 Find $\vec{a}_2 - \vec{a}_1$.

$\vec{a}_2 - \vec{a}_1 = 2\hat{i} + \hat{j} + \hat{k}$

4 Find $\vec{b} \times (\vec{a}_2 - \vec{a}_1)$. ✓ 1

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 6 \\ 2 & 1 & 1 \end{vmatrix} = -9\hat{i} + 10\hat{j} + 8\hat{k}$$

5 Find the two magnitudes. ✓ ½

$\sqrt{81 + 100 + 64} = \sqrt{245} = 7\sqrt{5}, \qquad |\vec{b}| = \sqrt{4 + 9 + 36} = 7$

6 Put into the formula. ✓ ½

$d = \dfrac{7\sqrt{5}}{7} = \sqrt{5}$ units

Answer **Both have DRs $2, -3, 6$; distance $= \sqrt{5}$ units**

Watch out: Taking DRs $4, -3, 12$ from the second line without first making the coefficients of x and z equal to 1.

11 **Short answer** NCERT Exemplar 2 marks

Find the angle between the lines whose direction cosines are given by the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$.

Formula / idea: For DCs l_1, m_1, n_1 and l_2, m_2, n_2: $\cos\theta = |l_1 l_2 + m_1 m_2 + n_1 n_2|$.

1 Put $n = -(l + m)$ in the second equation. ✓ ½

$l^2 + m^2 - (l + m)^2 = 0$

2 Expand and simplify. ✓ ½

$-2lm = 0 \;\Rightarrow\; l = 0$ or $m = 0$

3 Use $n = -(l + m)$ and $l^2 + m^2 + n^2 = 1$ in each case. ✓ ½

$l = 0 : \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

$m = 0 : \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$

4 Find the angle. ✓ ½

$\cos\theta = \left|0 + 0 + \dfrac{1}{2}\right| = \dfrac{1}{2} \;\Rightarrow\; \theta = \dfrac{\pi}{3}$

Answer $\dfrac{\pi}{3}$

Watch out: Forgetting $l^2 + m^2 + n^2 = 1$; the ratios alone are DRs, not DCs.

12 **Short answer** Board-pattern 2 marks

Find the value of a for which the lines $\vec{r} = (2\hat{i} - \hat{j}) + \lambda(a\hat{i} + 2\hat{j} - \hat{k})$ and $\frac{1-x}{3} = \frac{2y+1}{4} = z + 2$ are perpendicular.

Formula / idea: Perpendicular lines: $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

1	DRs of the first line from its direction vector. $a,\ 2,\ -1$	✓ ½
2	Write the second line in standard form. $\frac{x-1}{-3} = \frac{y+\frac{1}{2}}{2} = \frac{z+2}{1} \Rightarrow \text{DRs } -3, 2, 1$	✓ ½
3	Apply the perpendicular condition. $a(-3) + 2(2) + (-1)(1) = 0$	✓ ½
4	Solve. $-3a + 3 = 0 \Rightarrow a = 1$	✓ ½

Answer $a = 1$

Watch out: Taking DRs $3, 4, 1$ from $\frac{1-x}{3} = \frac{2y+1}{4}$; rewrite as $\frac{x-1}{-3}$ and $\frac{y+\frac{1}{2}}{2}$ first.

13 **Short answer** NCERT Ex 11.1 Q5 2 marks

Find the direction cosines of the sides of the triangle whose vertices are $A(3, 5, -4)$, $B(-1, 1, 2)$ and $C(-5, -5, -2)$.

Formula / idea: DCs of PQ: $\frac{x_2 - x_1}{PQ}, \frac{y_2 - y_1}{PQ}, \frac{z_2 - z_1}{PQ}$.

1	DRs and length of each side. $AB: -4, -4, 6;\ \sqrt{68} = 2\sqrt{17}$ $BC: -4, -6, -4;\ \sqrt{68} = 2\sqrt{17}$ $CA: 8, 10, -2;\ \sqrt{168} = 2\sqrt{42}$	✓ ½
2	DCs of AB. $-\frac{2}{\sqrt{17}}, -\frac{2}{\sqrt{17}}, \frac{3}{\sqrt{17}}$	✓ ½
3	DCs of BC. $-\frac{2}{\sqrt{17}}, -\frac{3}{\sqrt{17}}, -\frac{2}{\sqrt{17}}$	✓ ½

4 DCs of CA. ✓ ½
$\frac{4}{\sqrt{42}}, \frac{5}{\sqrt{42}}, -\frac{1}{\sqrt{42}}$

Answer AB: $-\frac{2}{\sqrt{17}}, -\frac{2}{\sqrt{17}}, \frac{3}{\sqrt{17}}$; BC: $-\frac{2}{\sqrt{17}}, -\frac{3}{\sqrt{17}}, -\frac{2}{\sqrt{17}}$; CA: $\frac{4}{\sqrt{42}}, \frac{5}{\sqrt{42}}, -\frac{1}{\sqrt{42}}$

Watch out: Dividing the DRs by their sum instead of by the length of the side.

14 **Short answer** Board-pattern 2 marks

Find the vector and the Cartesian equations of the line passing through the point $(2, -1, 5)$ and parallel to the line $4 - x = \frac{2y+1}{6} = \frac{3z-2}{-9}$.

Formula / idea: Parallel lines have proportional DRs; read DRs only after the coefficients of x, y, z are 1.

1 Write the given line in standard form. ✓ ½
$\frac{x-4}{-1} = \frac{y+\frac{1}{2}}{3} = \frac{z-\frac{2}{3}}{-3}$

2 Read the DRs; multiply by -1. ✓ ½
$-1, 3, -3 \Rightarrow 1, -3, 3$

3 Vector equation through $(2, -1, 5)$. ✓ ½
$\vec{r} = (2\hat{i} - \hat{j} + 5\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 3\hat{k})$

4 Cartesian equation. ✓ ½
$\frac{x-2}{1} = \frac{y+1}{-3} = \frac{z-5}{3}$

Answer $\vec{r} = (2\hat{i} - \hat{j} + 5\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 3\hat{k})$; $\frac{x-2}{1} = \frac{y+1}{-3} = \frac{z-5}{3}$

Watch out: Reading DRs $1, 6, -9$ from the given form; divide $2y + 1$ by 2 and $3z - 2$ by 3 first.

Level 3

27 marks

15 **Long answer** NCERT Exemplar 5 marks

Find the foot of the perpendicular from the point $(2, 3, -8)$ to the line $\frac{4-x}{2} = \frac{y}{6} = \frac{1-z}{3}$. Also, find the perpendicular distance from the given point to the line.

Formula / idea: The foot L lies on the line and $\overrightarrow{PL}$ is perpendicular to the line's direction.

1 Write the line in standard form. ✓ ½
$\frac{x-4}{-2} = \frac{y-0}{6} = \frac{z-1}{-3} \Rightarrow \text{DRs } -2, 6, -3$

2	Put each fraction equal to λ to get a general point L. $L(4-2\lambda,\ 6\lambda,\ 1-3\lambda)$	✓ 1
3	DRs of PL, where $P(2,3,-8)$. $2-2\lambda,\ 6\lambda-3,\ 9-3\lambda$	✓ ½
4	PL is perpendicular to the line. $-2(2-2\lambda)+6(6\lambda-3)-3(9-3\lambda)=0$	✓ ½
5	Solve for λ. $49\lambda-49=0\ \Rightarrow\ \lambda=1$	✓ ½
6	Put $\lambda=1$ in L. $L(2,\ 6,\ -2)$	✓ 1
7	Find the length PL. $PL=\sqrt{(2-2)^2+(6-3)^2+(-2+8)^2}=\sqrt{0+9+36}$	✓ ½
8	Hence the perpendicular distance. $PL=\sqrt{45}=3\sqrt{5}$ units	✓ ½

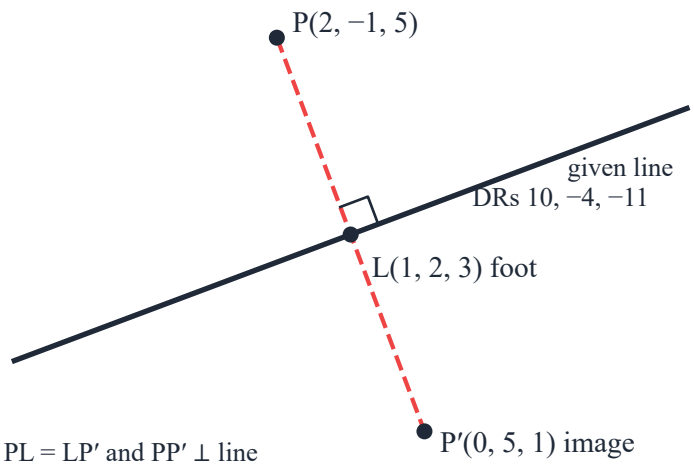
Answer **Foot $(2,6,-2)$; perpendicular distance $=3\sqrt{5}$ units**

Watch out: Using DRs $2,6,3$ from the given form; $4-x$ and $1-z$ flip the signs to -2 and -3.

16 **Long answer** PYQ-style 5 marks

Find the image P' of the point $P(2,-1,5)$ in the line $\vec{r}=(11\hat{i}-2\hat{j}-8\hat{k})+\lambda(10\hat{i}-4\hat{j}-11\hat{k})$. Also find the vector equation of the line joining P and P'.

Formula / idea: The foot L of the perpendicular from P is the mid-point of P and its image P'.



1	A general point L on the line. $L(11+10\lambda,\ -2-4\lambda,\ -8-11\lambda)$	✓ 1

2	DRs of PL. $9+10\lambda,\ -1-4\lambda,\ -13-11\lambda$	✓ ½
3	PL is perpendicular to the line (DRs $10, -4, -11$). $10(9+10\lambda)-4(-1-4\lambda)-11(-13-11\lambda)=0$	
4	Simplify and solve for λ. $237\lambda+237=0 \Rightarrow \lambda=-1$	✓ 1
5	Foot of the perpendicular ($\lambda=-1$). $L(1,\ 2,\ 3)$	✓ ½
6	L is the mid-point of PP', so $P'=2L-P$. $P'=(2-2,\ 4+1,\ 6-5)=(0,\ 5,\ 1)$	✓ 1
7	Direction of PP'; divide by -2. $(0-2,\ 5+1,\ 1-5)=(-2,6,-4) \Rightarrow 1,-3,2$	
8	Vector equation of PP' through P. $\vec{r}=(2\hat{i}-\hat{j}+5\hat{k})+\mu(\hat{i}-3\hat{j}+2\hat{k})$	✓ 1

Answer **Image $P'(0,5,1)$; line PP':** $\vec{r}=(2\hat{i}-\hat{j}+5\hat{k})+\mu(\hat{i}-3\hat{j}+2\hat{k})$

Watch out: Giving the foot $L(1,2,3)$ as the image; the image lies as far beyond L as P is before it.

17 **Long answer** CBSE 2023 5 marks

Find the equations of the diagonals of the parallelogram $PQRS$ whose vertices are $P(4,2,-6)$, $Q(5,-3,1)$, $R(12,4,5)$ and $S(11,9,-2)$. Use these equations to find the point of intersection of the diagonals.

Formula / idea: Two lines meet if their general points agree for some λ, μ in all three coordinates.

1	Diagonal PR: DRs $8, 2, 11$, through $P(4,2,-6)$. $\frac{x-4}{8}=\frac{y-2}{2}=\frac{z+6}{11}$	✓ 1
2	Diagonal QS: DRs $6, 12, -3$, i.e. $2, 4, -1$, through $Q(5,-3,1)$. $\frac{x-5}{2}=\frac{y+3}{4}=\frac{z-1}{-1}$	✓ 1
3	General points on the two diagonals. $(8\lambda+4,\ 2\lambda+2,\ 11\lambda-6), \qquad (2\mu+5,\ 4\mu-3,\ -\mu+1)$	✓ ½
4	Equate the x and y coordinates. $8\lambda-2\mu=1, \qquad 2\lambda-4\mu=-5$	
5	Solve the two equations. $\lambda=\frac{1}{2}, \quad \mu=\frac{3}{2}$	✓ 1

6 Check the z coordinates. ✓ ½

$11\left(\frac{1}{2}\right) - 6 = -\frac{1}{2} = -\frac{3}{2} + 1$ ✓

All three agree, so the diagonals intersect.

7 Put $\lambda = \frac{1}{2}$ in the first general point. ✓ 1

$\left(8,\ 3,\ -\frac{1}{2}\right)$

Answer PR: $\frac{x-4}{8} = \frac{y-2}{2} = \frac{z+6}{11}$; QS: $\frac{x-5}{2} = \frac{y+3}{4} = \frac{z-1}{-1}$; **they meet at** $\left(8, 3, -\frac{1}{2}\right)$

Watch out: Joining P to Q and R to S; those are sides. The diagonals are PR and QS.

18 **Long answer** NCERT Exemplar 5 marks

Find the equations of the two lines through the origin which intersect the line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ at angles of $\frac{\pi}{3}$ each.

Formula / idea: The required line joins O to a point P of the given line; $\cos\theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$.

1 A general point P on the given line. ✓ ½

$P(2\lambda + 3,\ \lambda + 3,\ \lambda)$

2 DRs of OP are the coordinates of P; the given line has DRs $2, 1, 1$. ✓ ½

$\vec{b}_1 = (2\lambda + 3,\ \lambda + 3,\ \lambda), \quad \vec{b}_2 = (2, 1, 1)$

3 Use $\cos\frac{\pi}{3} = \frac{1}{2}$. ✓ 1

$$\frac{|2(2\lambda+3) + (\lambda+3) + \lambda|}{\sqrt{6}\sqrt{(2\lambda+3)^2 + (\lambda+3)^2 + \lambda^2}} = \frac{1}{2}$$

4 Simplify the numerator and the square root.

$$\frac{|6\lambda + 9|}{\sqrt{6}\sqrt{6\lambda^2 + 18\lambda + 18}} = \frac{1}{2}$$

5 Square and cross-multiply. ✓ 1

$4(6\lambda + 9)^2 = 6(6\lambda^2 + 18\lambda + 18) \ \Rightarrow\ 108\lambda^2 + 324\lambda + 216 = 0$

6 Divide by 108 and factorise. ✓ ½

$\lambda^2 + 3\lambda + 2 = 0 \ \Rightarrow\ \lambda = -1$ or $\lambda = -2$

7 The two points of intersection. ✓ ½

$\lambda = -1 : P_1(1, 2, -1), \qquad \lambda = -2 : P_2(-1, 1, -2)$

8 Lines through the origin and P_1, P_2.
$\frac{x}{1}=\frac{y}{2}=\frac{z}{-1}$ and $\frac{x}{-1}=\frac{y}{1}=\frac{z}{-2}$ ✓ 1

Answer $\frac{x}{1}=\frac{y}{2}=\frac{z}{-1}$ **and** $\frac{x}{-1}=\frac{y}{1}=\frac{z}{-2}$

Watch out: Dropping the modulus and keeping only one root; both values of λ give a line at $\frac{\pi}{3}$.

19 **Short answer** CBSE SQP 2026-27 3 marks

Find the angle between the lines $\frac{x-1}{2}=\frac{y-1}{2}=\frac{z-2}{1}$ and $\frac{x-5}{2}=y-\frac{1}{2}=\frac{3-z}{2}$. Also find the Cartesian equation of a line passing through the point $(1,1,1)$ and perpendicular to both lines.

Formula / idea: A line perpendicular to both $\vec{b}_1$ and $\vec{b}_2$ is parallel to $\vec{b}_1\times\vec{b}_2$.

1 Write the second line in standard form.
$\frac{x-5}{2}=\frac{y-\frac{1}{2}}{1}=\frac{z-3}{-2}\Rightarrow$ DRs $2,1,-2$ ✓ ½

2 Angle between the lines (DRs $2,2,1$ and $2,1,-2$).
$\cos\theta=\frac{|2(2)+2(1)+1(-2)|}{\sqrt{9}\sqrt{9}}=\frac{4}{9}\Rightarrow\theta=\cos^{-1}\left(\frac{4}{9}\right)$ ✓ 1

3 Find $\vec{b}_1\times\vec{b}_2$.
$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&2&1\\2&1&-2\end{vmatrix}=-5\hat{i}+6\hat{j}-2\hat{k}$ ✓ 1

4 Line through $(1,1,1)$ with DRs $-5,6,-2$.
$\frac{x-1}{-5}=\frac{y-1}{6}=\frac{z-1}{-2}$ ✓ ½

Answer $\theta=\cos^{-1}\left(\frac{4}{9}\right)$; $\frac{x-1}{-5}=\frac{y-1}{6}=\frac{z-1}{-2}$

Watch out: Taking DRs $2,1,2$ for the second line; $3-z$ makes the DR of z equal to -2.

20 **Case study** PYQ-style 4 marks

Case Study: Two insects crawl along straight paths in a garden. The first insect moves along the line $l_1:\vec{r}=6\hat{i}+2\hat{j}+2\hat{k}+\lambda(\hat{i}-2\hat{j}+2\hat{k})$ and the second along the line $l_2:\vec{r}=-4\hat{i}-\hat{k}+\mu(3\hat{i}-2\hat{j}-2\hat{k})$. A biologist wants to know how close the two insects can ever get. Based on the given information, answer the following questions.

(i) Write the direction ratios of the two paths and show that the paths are not parallel. 1 mark

(ii) Find a vector perpendicular to both the paths. 1 mark

(iii) Find the shortest distance between the two paths. 2 marks

Formula / idea: Skew lines: $d = \left| \dfrac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$.

(i)

1 mark

1	Read the direction ratios of the two paths. $l_1 : 1, -2, 2, \qquad l_2 : 3, -2, -2$	✓ ½
2	Compare the ratios. $\dfrac{1}{3} \neq \dfrac{-2}{-2}$	✓ ½
3	The DRs are not proportional, so the paths are not parallel.	

Answer (i) $\dfrac{1}{3} \neq \dfrac{-2}{-2}$**, so** l_1 **and** l_2 **are not parallel.**

(ii)

1 mark

1	A vector perpendicular to both paths is $\vec{b}_1 \times \vec{b}_2$. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix} = 8\hat{i} + 8\hat{j} + 4\hat{k}$	✓ 1

Answer (ii) $8\hat{i} + 8\hat{j} + 4\hat{k}$ **(or any multiple, such as** $2\hat{i} + 2\hat{j} + \hat{k}$**)**

(iii)

2 marks

1	Find $\vec{a}_2 - \vec{a}_1$. $(-4\hat{i} - \hat{k}) - (6\hat{i} + 2\hat{j} + 2\hat{k}) = -10\hat{i} - 2\hat{j} - 3\hat{k}$	✓ ½
2	Find the dot product with $\vec{b}_1 \times \vec{b}_2$. $8(-10) + 8(-2) + 4(-3) = -108$	✓ ½
3	Find $\lvert\vec{b}_1 \times \vec{b}_2\rvert$. $\sqrt{64 + 64 + 16} = \sqrt{144} = 12$	✓ ½
4	Put into the shortest distance formula. $d = \left\lvert \dfrac{-108}{12} \right\rvert = 9$ units	✓ ½

Answer (iii) 9 **units**

Watch out: Writing the answer as -9; the shortest distance is the modulus, so it is 9 units.